# Handheld Adjustment Guide

01 Getting started with Handheld Zebra TC22

## Understand the screen. Follow each adjustment step.

Learn the Handheld workflow-from checking machine status and selecting a part and adjustment point to reading reports-with DENSO and CROBOTP examples.

## Check the device and information before opening adjustment controls

1) Check the Handheld Zebra TC22, its battery and screen. Use the wake/sleep button shown, then unlock the device using the method specified at your workplace.

2) Connect to an approved network, then open the installed Handheld Adjustment app or the shortcut provided by your administrator. If there is no shortcut, contact the administrator.

3) Wait for the Dashboard to load. Check SERVER, FLEET and UPDATED before reading the Robot cards. The 11/11 shown is the result at the time of the photograph, not a condition guaranteed at all times.

4) Match the Robot number on the physical machine to the card on the screen. Before tapping OPEN ADJUST, check the Part and your assigned job again.

### Before continuing

ONLINE indicates connectivity; it does not certify that the area is safe. In an emergency, use the emergency-stop system and your workplace emergency response procedure. The web screen is not an emergency-stop device.

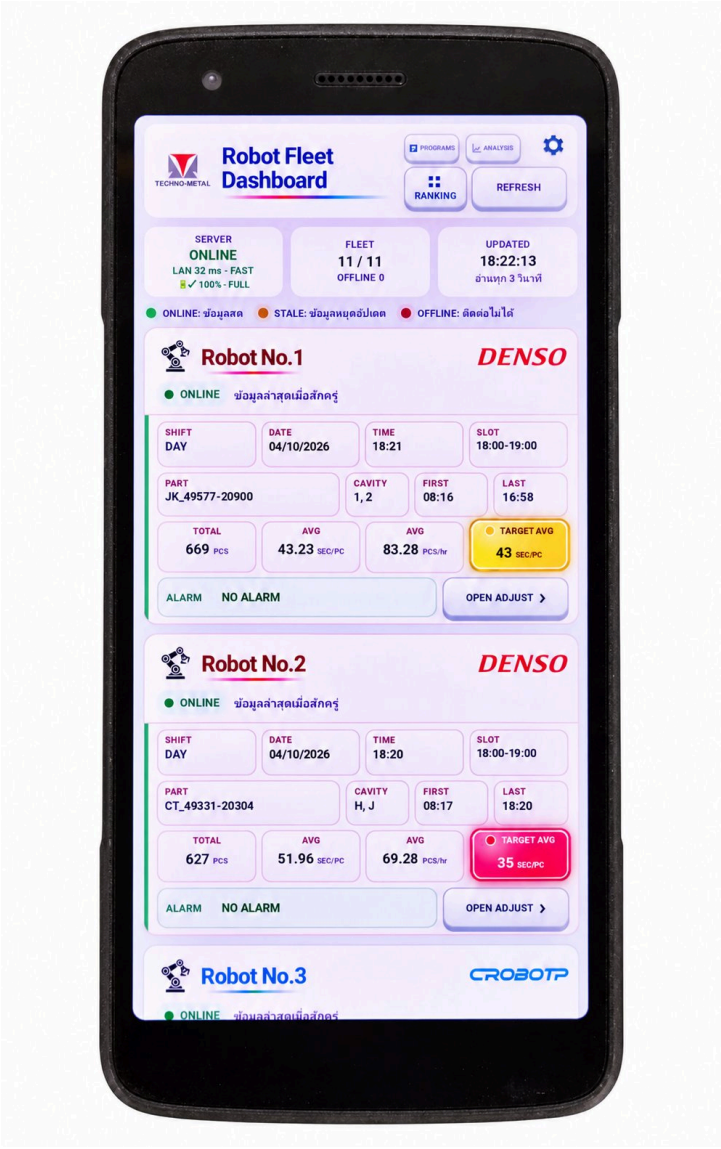



1 · Handheld in screen-sleep mode, showing the clock and device battery level

# 01 Getting started with Handheld Zebra TC22



2 · Physical Power button for powering the device on/off and waking/ sleeping the screen




3 · Handheld displaying Mobile Adjust at the Robot Fleet Dashboard

# Handheld Adjustment Guide

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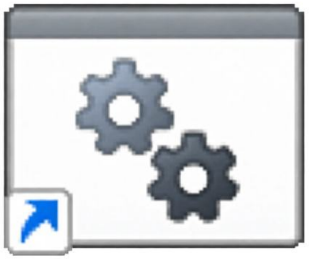
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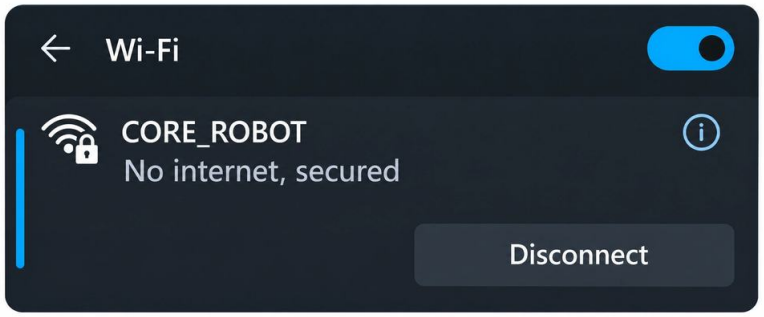
33 Simulator 45

# 02 Checks when the Server cannot be reached



**START_SERVER**

START_SERVER shortcut icon on the Laptop Server.




Windows Wi-Fi example: CORE_ROBOT - No internet, secured.

Initial checks when the Handheld cannot reach the Server-not a guaranteed fix for every cause. These are Windows Laptop Server references: START_SERVER and Wi-Fi, not TC22 Wi-Fi settings.

Caution: No internet means Windows has not confirmed internet access; it does not by itself prove the local network or server is unavailable. Wi-Fi association does not prove server readiness either. If data does not arrive, check the network, Laptop and application under the WI and contact the administrator. Do not disable security controls or use Robot Emergency/Reset to fix network issues.

1) Check that the Laptop is powered and connected to the workplace-designated network. The example is connected to CORE_ROBOT and shows No internet, secured. Disconnect ends the network connection; it does not start the server.

2) Use the administrator's prescribed method to check whether the Server application is already running. If it is, do not start another copy; check its status or contact the administrator. If it is not running, use START_SERVER under the WI. The icon identifies the shortcut but does not confirm successful startup.

3) Connect the Handheld to a network configured to reach the Laptop Server, then open Handheld Adjustment through the approved app/shortcut. Do not guess an IP address or URL.

4) Once the Dashboard loads, check SERVER, FLEET and UPDATED for fresh responses and match the Robot/Part to the job. A page that opens without updating its timestamp does not establish readiness.

# 03-04 Before starting and shared controls

## 03 Before starting

### Robot · Part · Jig · Line

Verify robot/brand, confirmed part, program, jig and line. P/LP ranges are profile-specific, not interchangeable across machines.

### UPDATED

Check UPDATED and data age.

### Check scope before confirming

Before Apply/Confirm/Recall recheck axis, sign and scope. Undo is scoped, not guaranteed full-state restoration.

### On a fault: request a status reread under the site WI

Site-specific procedure for trained, authorized personnel only, on machines and cases whose approved WI specifies this sequence: 1) Press Emergency. 2) Check that the area is safe and the cause has been addressed before releasing Emergency. 3) Press Preparation as specified by the WI. 4) Press Reset twice to request a Controller status reread. Then verify fresh data, no Alarm and the readiness permitted by the WI. Release/Reset does not prove recovery or authorize motion. If the fault remains, stop and contact the responsible person; do not repeat the reset cycle. This differs from CALL acknowledgement and DENSO Recovery, and is not a general remedy for Wi-Fi, stale data or battery issues.

## 04 Shared controls

### Navigation

Tabs change views; SELECT ROBOT changes robot; X closes; search/ arrows navigate lists; image buttons open references, not robot motion.

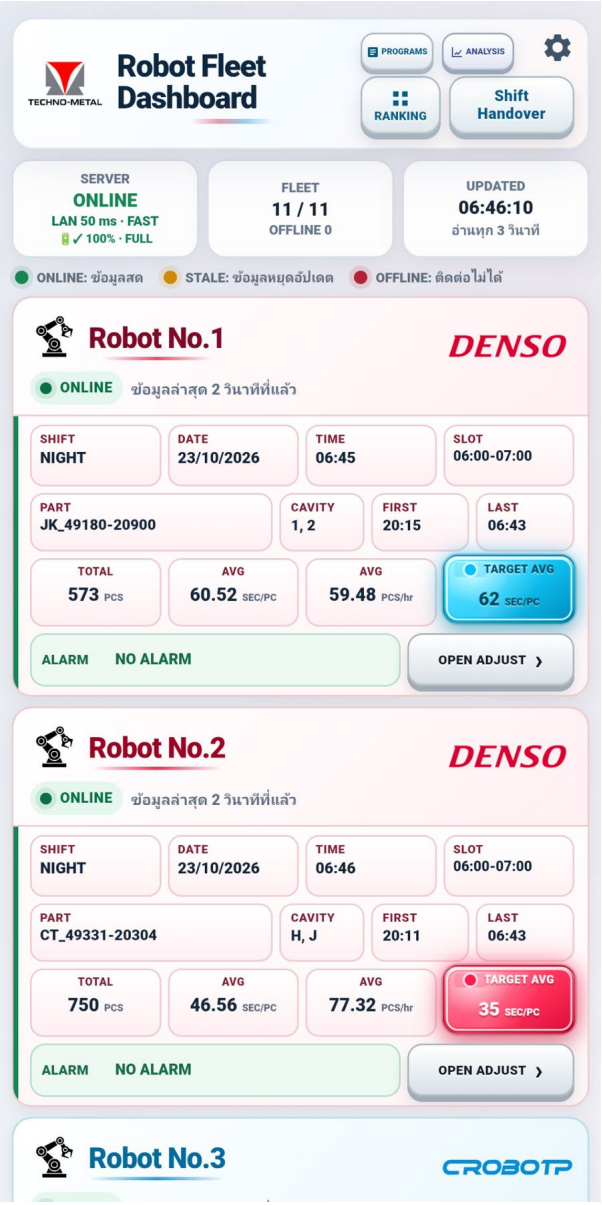
### Report / PNG / PDF

DAY/NIGHT selects report shift; PNG downloads that report; the language-adjacent PDF is this landscape A4 manual. EXTEND SHIFT changes shared aggregation: DAY through22:00, NIGHT from18:00; OFF restores STANDARD, not machine operation.

### History

ALL filters category; TODAY/7D/30D filter time; Search matches text; REFRESH updates. In DENSO, AVAILABLE is the calendar span from the first to last filtered record, not a distinct-day count.

# 05 Fleet Dashboard: fleet overview




Start with data freshness, then read robot cards and choose a destination.

## Using and reading this screen

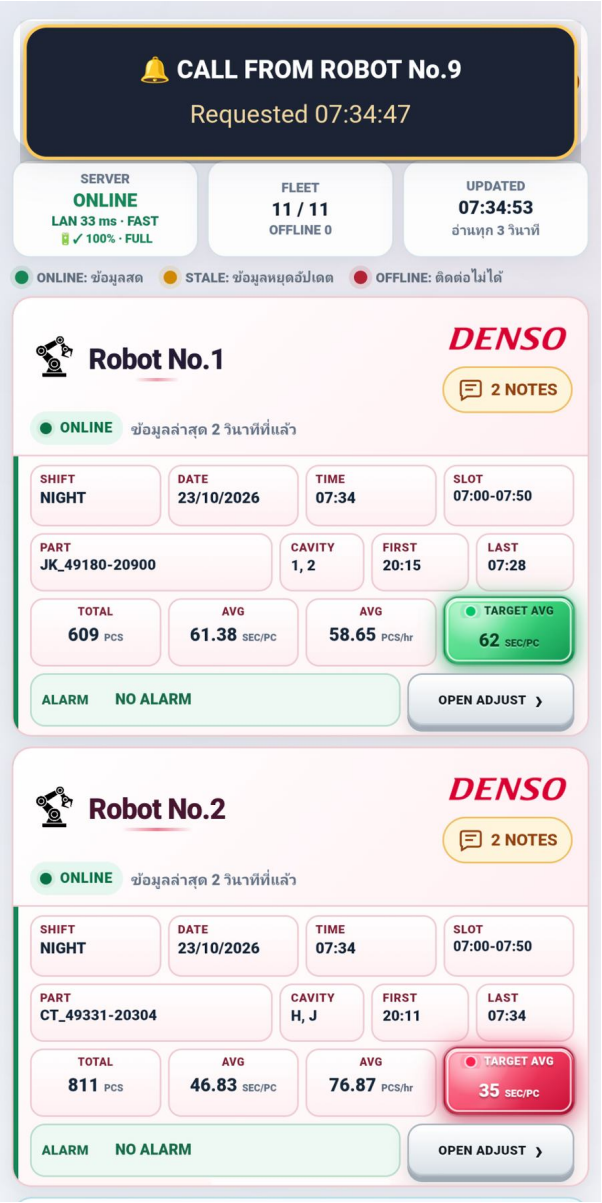
1) Check SERVER ONLINE, LAN, FLEET 11/11, OFFLINE 0 and UPDATED 06:46:10; the legend distinguishes ONLINE/STALE/OFFLINE.

2) On each card, read shift, date, time, slot, part, cavity and first/last before totals, averages and target; their units differ.

3) PROGRAMS opens availability, ANALYSIS analysis, RANKING overview, Shift Handover notes and the gear settings; OPEN ADJUST belongs to its robot card.

Caution: FLEET11/11 indicates coverage, not that every machine is producing.

# 06 Robot call banner




Read the calling robot number before opening adjustment; it may differ from the cards underneath.

## Using and reading this screen

1) CALL FROM ROBOT No.9 identifies the caller and Requested 07:34:47, not a motion instruction.

2) Check SERVER, FLEET and UPDATED: the example shows 11/11 and 07:34:53. Check each card's data age separately.

3) Use OPEN ADJUST only on the intended robot card; 2 NOTES refers to that card's handover notes.

Caution: The caller may differ from the card beneath; Requested is not acknowledgment time.

# 07 CALL and Laptop Server battery




Robot No.2 cabinet reference: yellow RESET is between JIG A GO BACK and JIG B GO BACK, above the red EMERGENCY STOP. Verify the actual button name and Robot number before use.

Identify the Robot, acknowledge the call at its cabinet and verify alerts against fresh data.

Caution: Use RESET only under the approved workplace procedure: the same button may reset Robot faults. A CALL does not establish that the area or machine is ready.

## Calling the responsible person and stopping the call sound

Caution: Software creates a CALL after a continuous press reaches 3 seconds. With an existing CALL, a new accepted press clears it. Continuing the same hold does not create another CALL, but releasing and holding again may.

1) When that Robot has no active CALL and assistance is needed, use its yellow cabinet RESET under the approved procedure. Hold it so the system reads a continuous press for at least 3 seconds, then release. Verify CALL FROM ROBOT No. and the correct number on the Handheld.

2) On receiving a CALL, read the Robot number and Requested time, then inspect that Robot under the workplace safety procedure.

3) After accepting the call and checking the site, verify that this Robot's CALL remains active. Press its yellow RESET again under the approved procedure, then release. This new accepted press clears the existing CALL without waiting for the 3-second creation threshold.

4) Wait for Server confirmation and verify that this Robot's CALL card disappears. Sound may continue if another Robot still has a CALL. Do not press again after it disappears: a new held press could create another CALL.

# 07 CALL and Laptop Server battery (continued)




Robot No.2 cabinet reference: yellow RESET is between JIG A GO BACK and JIG B GO BACK, above the red EMERGENCY STOP. Verify the actual button name and Robot number before use.

5) If Wi-Fi/Server disconnects, an old CALL may remain displayed. Check connectivity and contact the responsible person. Silence or closing the receiver does not mean the Server CALL has cleared.

### When Laptop Server battery is low

1) If the Handheld displays or sounds a low Laptop Server battery alert, confirm that it concerns the Laptop battery, not the Handheld battery.

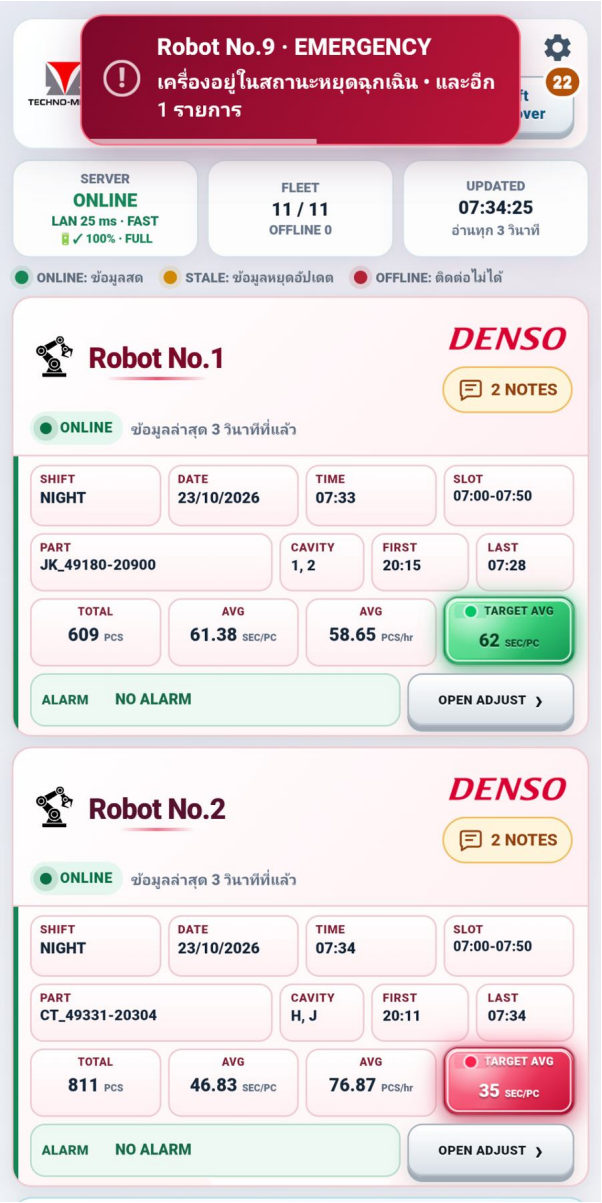
2) Connect the Laptop immediately to the workplace-approved power source and verify that the system confirms charging/AC connection.

3) For the alert configuration described in this guide, with fresh data showing battery operation, alerts repeat at 20-16% every 60 seconds; 15-11% every 30 seconds; 10-6% every 15 seconds; 5-0% every 5 seconds. The battery siren differs from CALL/Alarm sound.

4) Battery sound stops when AC is confirmed or charge exceeds 20%, but LOW BATTERY may remain until charge exceeds 20%. Without fresh data, battery sound is suppressed: check connectivity rather than assuming charging succeeded.

5) If it still runs on battery after connection or fresh data is unavailable, contact the responsible person. Do not use a Robot cabinet RESET to address Laptop battery issues.

# 08 Emergency banner




The example places a Robot No.9 emergency alert above the dashboard.

### Using and reading this screen

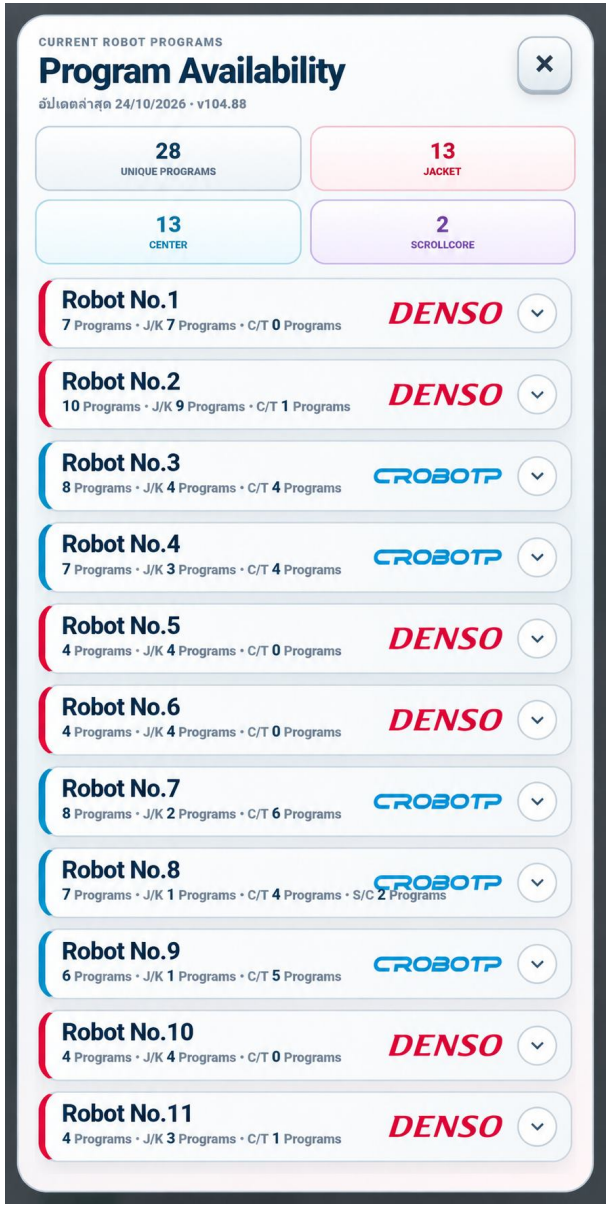
1) Read the robot number and EMERGENCY text; “one more item” means the banner is not the full alert list.

2) NO ALARM on No.1 or No.2 does not cancel the alert for No.9; interpret each robot separately.

3) For a real event, follow the site emergency plan and notify the responsible operator. Dismissing a banner does not fix its cause.

Caution: NO ALARM on No.1/2 does not clear No.9’s emergency; dismissing text does not fix the cause.

# 09 Program Availability




Use this page to inspect programs by robot and family, not to load or run them.

## Using and reading this screen

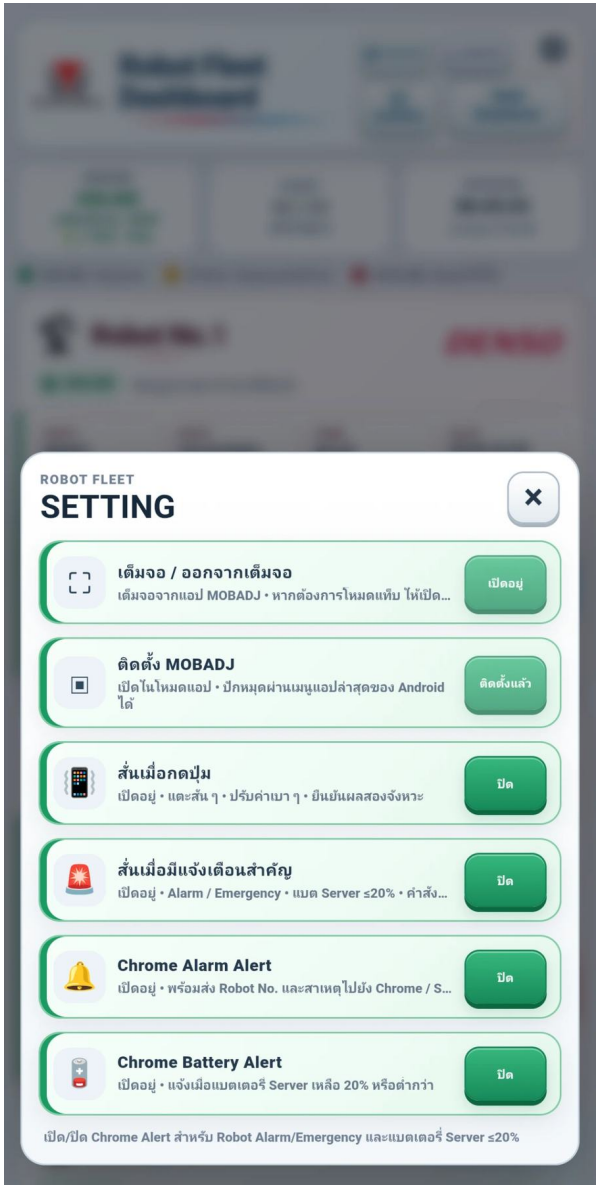
1) Read the update date and intended robot before counts or expanding program details.

2) The summary shows 28 unique programs: 13 Jacket, 13 Center, 2 Scrollcore. Unique totals are not the sum of all per-robot counts.

3) Expand a robot's chevron to inspect its list; check brand and family counts before choosing a part on the appropriate page.

Caution: A listed program is not necessarily ready to run; machine readiness is a separate check.

# 10 Fleet settings: display and alerts




Use this page to set Handheld display, vibration and notifications. Change only the item you need.

## Using and reading this screen

1) Open Setting and read the row name and current state before using its right-hand button.

2) If the description says enabled and the button says Turn off, the feature is currently on; tapping turns it off. To enable it again, tap Turn on and check the resulting state.

3) Tap X to close the window and return to Fleet. Closing the window does not turn notification settings off.

## Fullscreen and app installation

Fullscreen provides more space for information. Installing the app lets you open Handheld Adjustment from a shortcut; it does not start a Robot.

1) Use Fullscreen / Exit fullscreen to change the display. In app mode, enabled indicates that fullscreen is already active.

2) If Install MOBADJ says installed, do not install again. If an install button is available, tap it and follow the device prompts.

# 10 Fleet settings: display and alerts (continued)

### Button vibration versus alert vibration

These are separate settings. Turning off button vibration does not automatically turn off alert vibration.

1) Button vibration provides touch feedback, such as a short response to a tap and confirmation feedback.

2) Important-alert vibration is for events such as Alarm/Emergency and low Server battery. Enable or disable it according to workplace use.

### Chrome Alarm Alert and Chrome Battery Alert

These enable or disable Chrome notifications. They do not clear an Alarm or fix a battery problem.

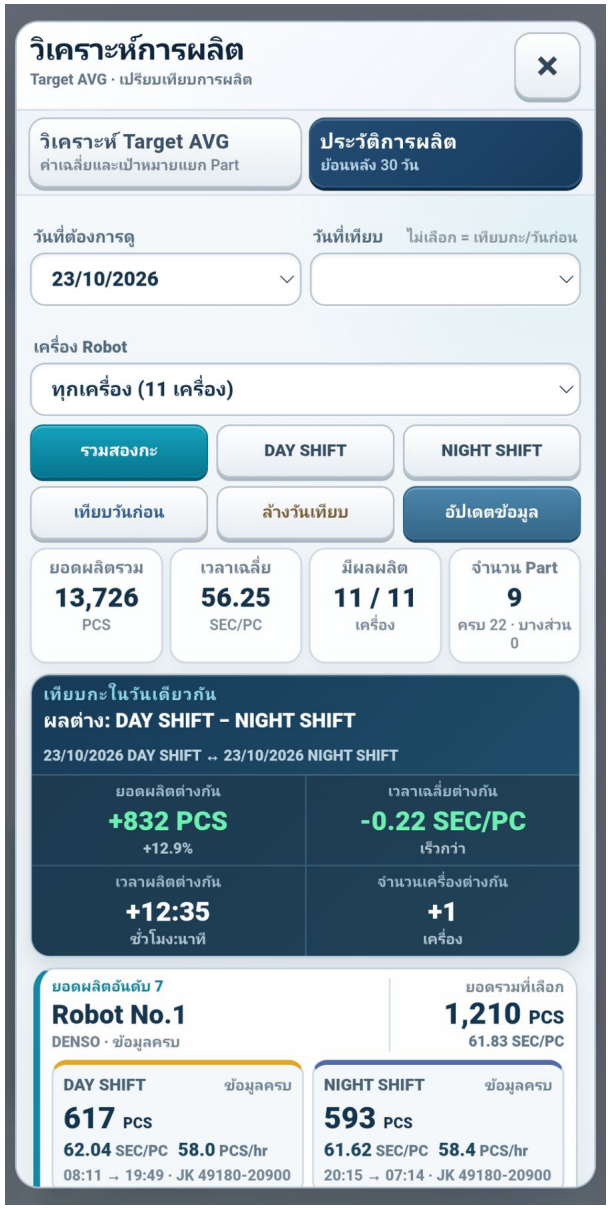
1) Chrome Alarm Alert identifies the Robot with Alarm/Emergency and its cause, helping you attend the correct machine.

2) Chrome Battery Alert warns at20% or below on the Laptop Server battery, not the Handheld. Check and connect the Laptop to power under the workplace procedure.

3) The same event may not produce another popup. Check the latest Fleet state to see whether it remains or has cleared; silence does not prove resolution.

Caution: Disabling notifications only stops that device notification; it does not clear an Alarm or restore Server battery charge.

# 11 Production history and shift comparison




Set the date, robot and shift scope before interpreting totals and differences.

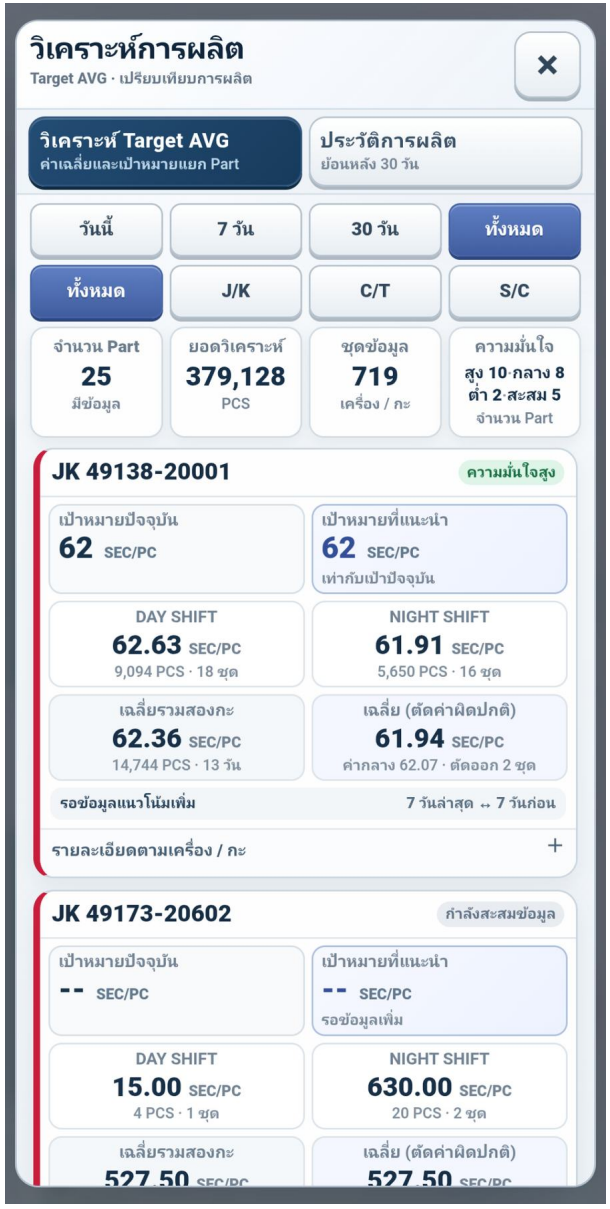
## Using and reading this screen

1) In Production History, choose the date and robot scope; this example uses 23/10/2026 and all 11 robots.

2) Choose both shifts, DAY SHIFT or NIGHT SHIFT. A blank comparison date uses the stated previous shift/day behavior; Clear comparison date removes that selection.

3) After setting filters, use Update Data. Read 13,726 PCS and 56.25 SEC/PC alongside 22 complete sets; the shown difference is DAY SHIFT minus NIGHT SHIFT.

# 12 Target AVG analysis




Compare observed averages and targets by part, with the data window and confidence level.

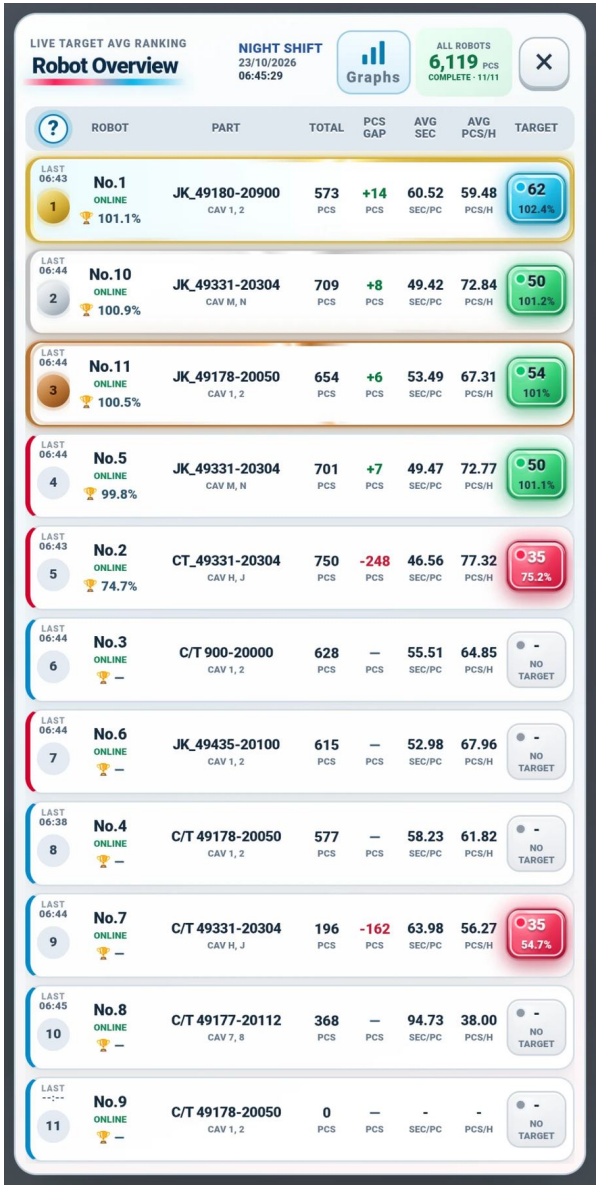
## Using and reading this screen

1) Select Today / 7 days / 30 days / All and J/K, C/T or S/C before reading part and sample totals.

2) JK 49138-20001 shows current and suggested targets of 62 SEC/PC, with separate day, night, combined and trimmed averages.

3) Use + for supporting detail. Treat “collecting data” and target -- as unavailable values, not zero.

# 13 Robot Overview: ranking and metrics




Read ranking with part and units; do not compare PCS/hr directly with SEC/PC.

## Using and reading this screen

1) Open Ranking and check the date, shift and data time. Tap Graphs to view the current Ranking data as charts. Counts and time may change when data refreshes.

2) Read each row's robot, part, total, gap, averages and target. -- or NO TARGET is not zero.

3) Graphs opens the chart view, ? is help and X closes the dialog. Viewing data does not change machine targets.

## 1. Rank uses the whole shift, not one Part's speed

SHIFT SCORE uses both output and elapsed work time: multiply pieces by Target AVG to express output as target-time seconds, then divide by elapsed work seconds and multiply by100. Both times must use seconds.

1) Suppose one Part has Target AVG =50 seconds/piece and60 pieces have been produced.

2) 60 minutes of scheduled work have elapsed, excluding scheduled breaks:60 ×60 =3,600 seconds.

3) Convert60 pieces into target-time output:60 ×50 =3,000 seconds.

4) Score:3,000 ÷3,600 ×100 =83.3%. The3,600 elapsed work seconds are the denominator: this is not a piece-count-only calculation.

5) Alternatively,3,600 seconds at50 seconds/piece gives a target of3,600 ÷50 =72 pieces. Producing60 of72 gives83.3%; producing72 gives100%.

# 13 Robot Overview: ranking and metrics (continued)

Verify: SPEED = Target AVG ÷ AVG SEC/PC ×100 describes the displayed Part, not the shift score. Card TOTAL/AVG/TARGET/PCS GAP may have a different scope from the whole-shift score.

### 2. How time and Part changes affect the score

The denominator starts at the scheduled shift start, not the first piece or a Part change. Scheduled breaks are excluded, but stops and changeovers within work periods still count. EXTEND adds scheduled work periods.

1) Continuing the Target50 example: after120 work minutes =7,200 seconds, if output is still60 pieces, the score is(60 ×50) ÷7,200 ×100 =41.7%. Time increased but output did not.

2) If output instead reaches120 pieces after120 work minutes, the score is(120 ×50) ÷7,200 ×100 =83.3%. Always read the quantity together with elapsed time.

3) A Part change does not remove earlier output. Use each Part's own Target: Part A60 pieces ×50 seconds + Part B20 pieces ×60 seconds =4,200 seconds. After120 work minutes =7,200 seconds,4,200 ÷7,200 ×100 =58.3%. Do not restart the elapsed time on a Part change.

4) Selecting a new Part without production adds no output. Alarm does not impose a fixed points penalty. Time stops at the final work period after shift end; NIGHT after midnight retains the shift-start date.

Verify: A - score means a verified score is unavailable, not0%; for example, data is incomplete or a produced Part lacks a Target. Do not substitute another Part's Target.

# 13 Robot Overview: ranking and metrics (continued)

### 3. Ordering and tie-breaks

Within the same group with whole-shift production, sort by displayed SHIFT SCORE rounded to one decimal, highest first-not primarily by SPEED or PCS GAP.

1) Tied score: higher whole-shift PCS across all Parts first; if still tied, lower Robot number first.

2) Production but score -: after scored machines in the same group, ordered by PCS descending.

3) Zero or missing whole-shift PCS moves to the end of the entire list, even with a configured Target.

Verify: Gold/silver/bronze mark ranks1/2/3. Target-light colour describes Part speed and need not match rank colour.

### 4. Connectivity and unverified report data

ONLINE/STALE with usable Reports receive rank numbers. OFFLINE with retained data is a separate group without rank numbers; WAITING/CHECK REPORT awaits verification.

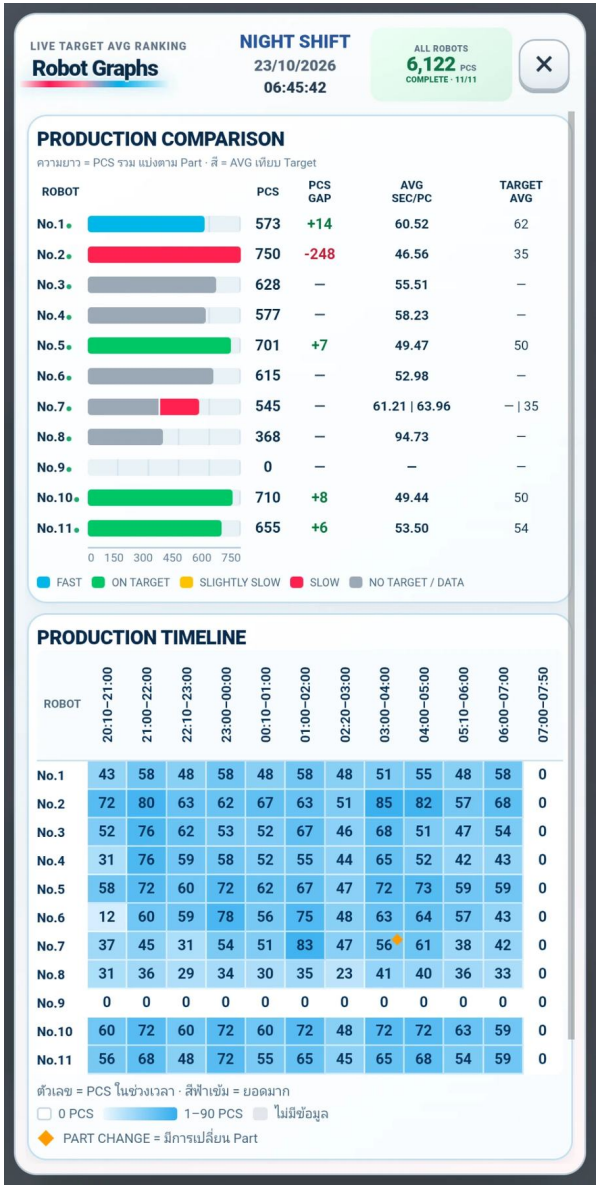
1) Among rows with output: ONLINE/STALE first, then OFFLINE using the same score rules, then waiting rows by Robot number; finally move all no-output rows to the end.

2) A - score does not always mean no rank number. Also check connectivity and Report quality.

3) Handheld Adjustment’s All 11 image download sorts scores without separating connectivity groups, so its order can differ from the screen.

Verify: ONLINE means data is being received, not proof of production. Read LAST, counts and freshness together; do not replace missing data with0.

# 14 Robot Graphs: comparison and timeline




Graphs presents the current Ranking data as charts and a timeline, showing which Robots and time periods contributed output.

## Using and reading this screen

1) From Ranking, tap Graphs and confirm the date and shift. Graphs uses the same scope, but refreshes can advance counts and time.

2) Use PRODUCTION COMPARISON to compare Robot totals and speed, then PRODUCTION TIMELINE to find periods with higher, lower or no recorded output.

3) Read the Part and values for each row, especially after a Part change during the shift. Tap X to close.

## 1. Read output and average time

Longer bars mean more PCS. Multiple Parts split the bar into segments. Bar colour compares speed with Target; it is not the ranking medal colour.

1) PCS is the displayed piece count for that Robot. PCS GAP compares actual pieces with expected pieces using recorded production time and each Part's Target, not whole-shift scheduled time: + above, − below; - means unavailable, not zero.

2) AVG SEC/PC is average time per piece: lower is faster. TARGET AVG is target time per piece, also in seconds.

3) For example, AVG60.52 versus TARGET62 is1.48 seconds faster per piece, so it is FAST. If values are separated by |, read them for their respective Part segments; do not add them together.

# 14 Robot Graphs: comparison and timeline (continued)

### 2. PCS GAP: how many pieces above or below target

PCS GAP = actual pieces − (recorded production seconds ÷ Target AVG). The result is a piece count, not a percentage or scrap count. This uses the production duration of the records being viewed, not the elapsed scheduled work time used in SHIFT SCORE.

1) Example: a Part's production records contain3,600 seconds with Target AVG50 seconds/piece. Expected output for this duration is3,600 ÷50 =72 pieces.

2) If actual output is75, PCS GAP =75 −72 =+3, shown in green:3 pieces above target. If actual output is60,60 −72 =−12, shown in red:12 below target.

3) Actual output72 gives0: on target for the recorded duration. The display rounds GAP to whole pieces, so a small difference can also round to0. A - means unavailable, for example because Target or data is incomplete. It is not zero, although both use grey.

4) For multiple Parts, calculate each difference using that Part's time and Target, add the differences, then round to whole pieces. Incomplete or inseparable Part data may show -. GAP is not the main ranking criterion.

### 3. Read colours against the target

Compare AVG SEC/PC with TARGET AVG for the same Part. These colours do not diagnose an Alarm or replace SHIFT SCORE.

# 14 Robot Graphs: comparison and timeline (continued)

1) Blue FAST: more than1 second/piece faster than target. Green ON TARGET: equal to target or up to1 second/piece faster.
2) Yellow SLIGHTLY SLOW: slower by less than1 second/piece. Red SLOW: slower by1 second/piece or more.
3) Grey NO TARGET / DATA: target or sufficient comparison data is missing. Speed cannot yet be classified; it does not mean zero output.

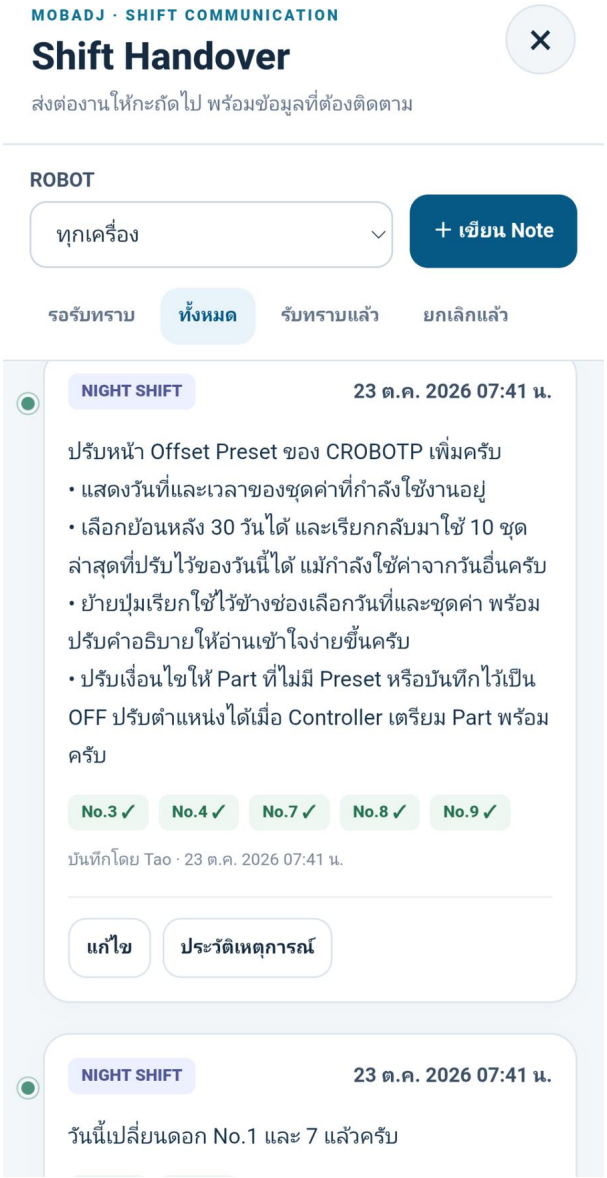
**4. Locate output by time period**

In PRODUCTION TIMELINE, rows are Robots, columns are time periods and cells show PCS for that period. Darker blue means more output, not the speed colours used above.

1) A white cell showing0 means no output is recorded for that period. Check its time heading: the period may not have started yet. Do not immediately infer a fault.
2) Grey “No data” means the count cannot be confirmed from available data. Check updates or the Report first; do not substitute0.
3) An orange PART CHANGE diamond marks a Part change in that period. Use it to interpret counts and averages when the old and new Parts have different cycle times.

Caution: 0 PCS is a displayed zero count; “No data” means the count is unconfirmed. If Graphs and Ranking differ, check date, shift, Part and refresh time first. A number or colour alone does not establish that a machine has stopped.

# 15 Shift Handover notes




Notes share shift context, separating robot scope, acknowledgment state, author and time.

## Using and reading this screen

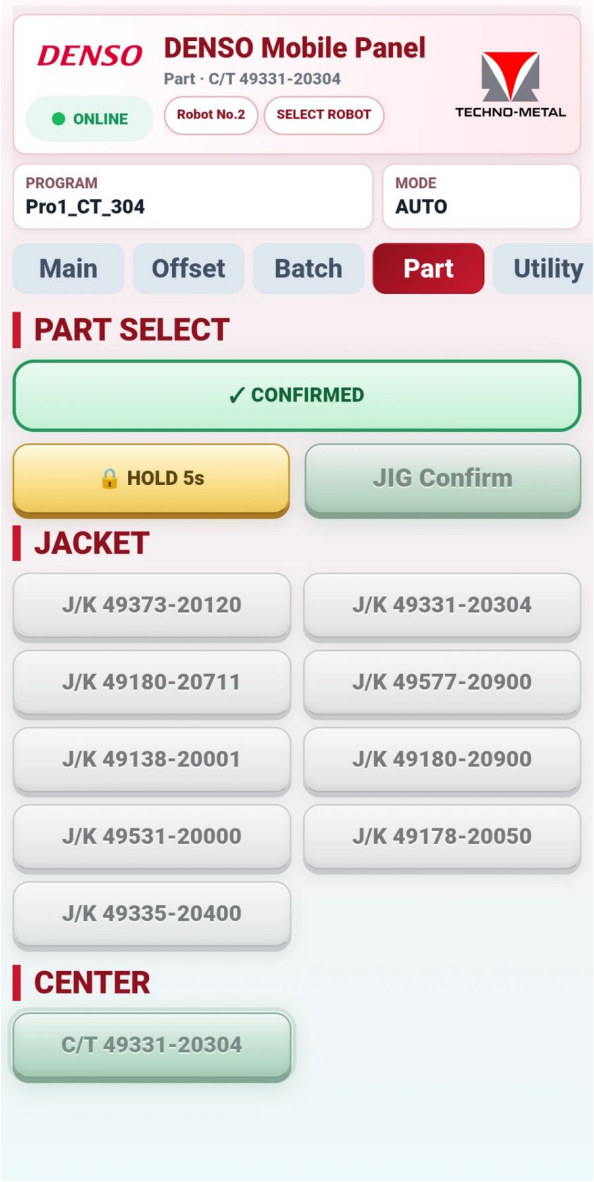
1) Use ROBOT to filter machines and Pending / All / Acknowledged / Canceled tabs to separate note states.

2) Read author Tao, NIGHT SHIFT, 23 Oct 2026 07:41 and robot references. A note's claims are not independent technical validation.

3) + Write Note starts a new note; Edit and Event History belong to existing notes. Training-note experiments remain inside this page.

Caution: Acknowledging a note does not certify repair or safety completion.

# 16 DENSO: part and jig confirmation




Select the Part to use, then press JIG Confirm to confirm it with the machine. These are two separate steps.

## Using and reading this screen

1) Check the Robot and complete Part code under Jacket/Center against the job.

2) In AUTO, hold HOLD5s fully to request temporary Part-selection unlocking for up to about 90 seconds. Early release cancels; unlock again if the window expires.

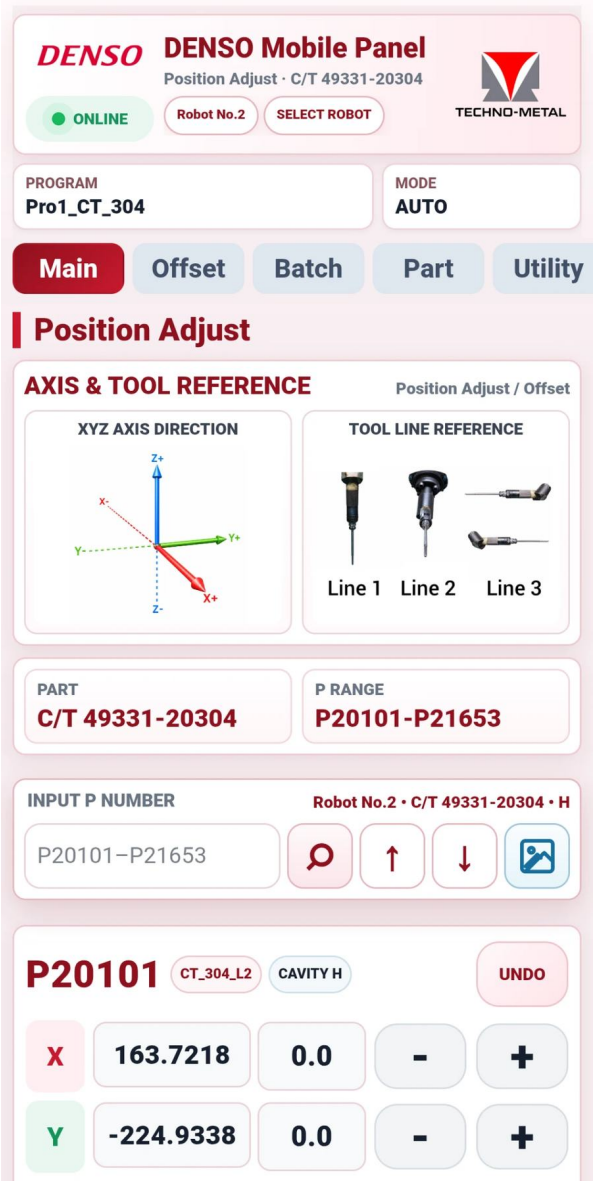
3) Tap the required Part and wait for the selection result. This submits the Part selection, not merely a preview. After successful selection, verify the selected Part and READY TO CONFIRM.

4) Check the Part and Jig against the job, then press JIG Confirm separately. Wait through CONFIRMING... for ✓ CONFIRMED. A successful response to Part selection is not Jig confirmation.

5) Confirmation in AUTO consumes the temporary unlock. On Error or Timeout, inspect selected and confirmed Parts before another request.

Caution: Select the Part to use. Check that the Part code and Jig match the job, then press JIG Confirm. Wait for ✓ CONFIRMED with the correct Part name to complete confirmation. If the name differs or confirmation is missing, check before adjusting.

# 17 DENSO: Main / Position Adjust




Select the job's P point and adjust its X/Y/Z axes individually. The steps below separate finding a point from writing values and verifying the result.

### 1. Confirm the job before selecting a point

Main / Position Adjust changes individual P positions for the selected Part. First match the Robot, CONFIRMED Part and Jig/Cavity to the actual job.

1) If the Part or context differs, follow the approved Part selection/confirmation procedure first; do not substitute a similar Part.

2) Use the image button to identify the point and Cavity base. A workpiece image does not establish an axis's +/− direction.

Verify: The context and intended point are confirmed. The P14833 example in the next section is not a universal point number.

### 2. Find P and check its card

Enter the complete P number in INPUT P NUMBER and tap ⌕ or Enter. ↑/↓ goes to the range's first/last point. Search displays the card; it does not command Robot motion.

1) Check the P number, program label and Cavity on the card before touching an axis control.

2) If the point is unavailable or outside the range, check the Part/P range; do not guess another number to bypass the condition.

Verify: The intended point's card is visible; screen position alone is not point confirmation.

### 3. Distinguish position values from Delta

Each X/Y/Z row shows a position value and a separate Delta. Delta is the change reported by the system-not a P number and not necessarily only the most recent tap.

1) Record X/Y/Z before the adjustment and check the approved axis, sign and amount.

Verify: All three pre-adjustment values are known for comparison with later readback.

# 17 DENSO: Main / Position Adjust (continued)

### 4. Adjust axes and verify confirmation

The card's axis −/+ sends a change to that P. Each press changes the position value by 0.1 mm; this is a distance increment per press. No extra APPLY/OK is required, unlike the prepared Delta in Batch.

1) Check P, Part, Cavity, axes and signs, then press −/+ for the required amount. Consecutive presses and changes to multiple axes are supported through a request queue; this does not mean the Controller executes all axes simultaneously.

2) After the intended presses, wait for the queue to finish. The CONFIRMED notice identifies the final request's P/axis/sign. Then check returned X/Y/Z against the requested changes; card values may update before completion.

3) If POSITION QUEUE FULL appears, wait for earlier requests to confirm before adding more. An Error clears remaining queued requests; inspect readback to establish which changes completed before starting again.

Verify: 0.1 mm is the distance change per press. After the queue finishes, verify CONFIRMED and readback against the requested changes.

### 5. UNDO and uncertain results

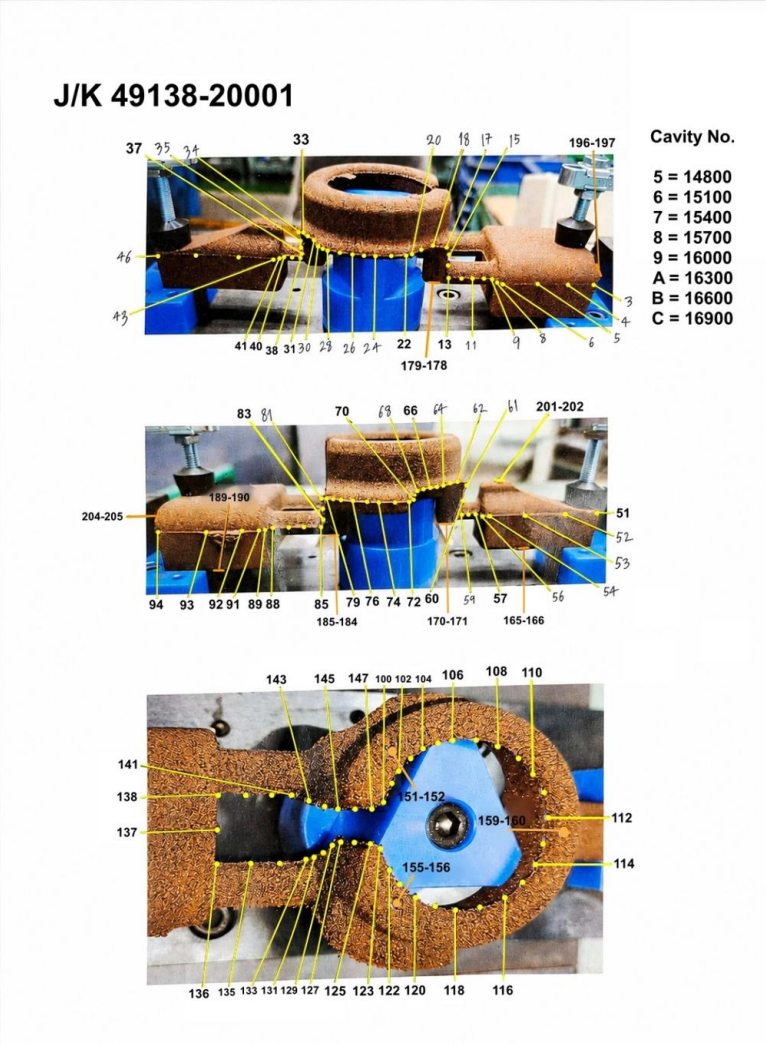
UNDO on a card requests restoration for that P; it does not merely discard an on-screen edit. Returned values include X/Y/Z, so check all three axes.

1) Before UNDO, check the same P/Part/Cavity and wait for prior commands to finish. Then wait for confirmation and verify readback.

2) On Error, Timeout or mismatched values, stop, read the actual state/values and contact the responsible person. Do not retry to guess whether the previous write succeeded.

Verify: Confirm the result from readback; do not assume UNDO can restore every historical change or context.

# 18 DENSO example: image point 33 → P14833




J/K 49138-20001 reference: number 33 in the top image has a yellow line pointing to the left shoulder of the central hole. The right table gives Cavity 5 = 14800, hence P14833 for Cavity 5 / Jig A.

Read the image, select the P number and verify readback for the approved point.

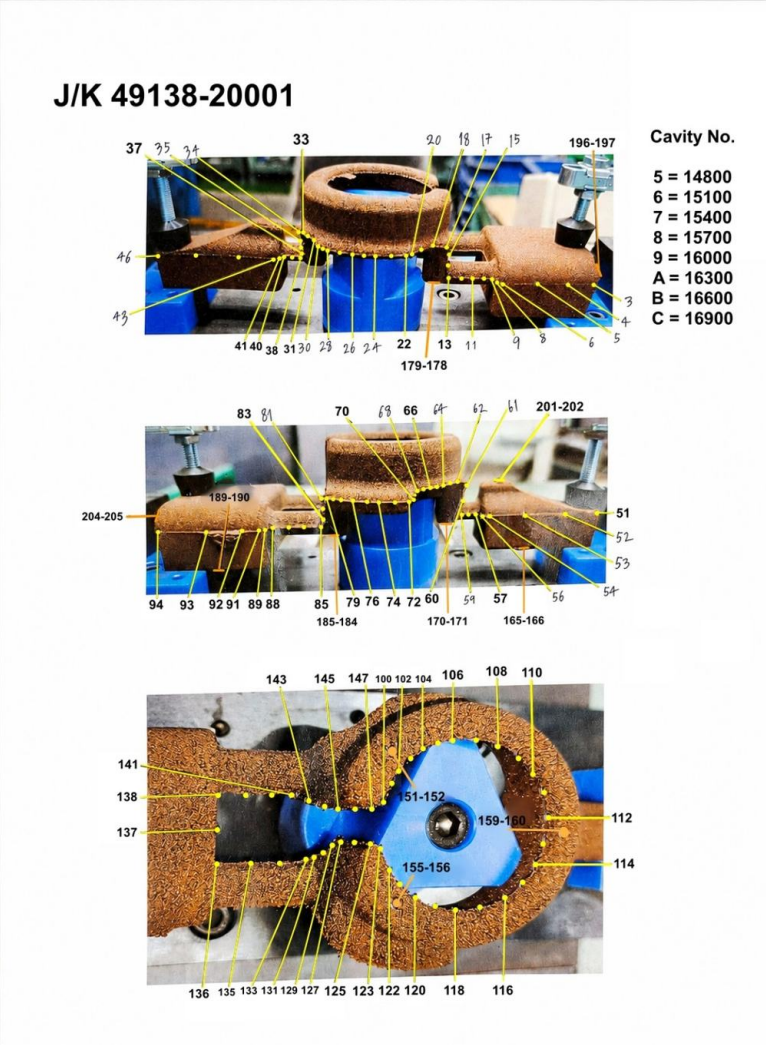
Caution: Match Robot, Part, Jig/Cavity, point, axes and signs to the job and follow the workplace WI. If offline or data is inconsistent, check status before adjusting. The workpiece image locates points; it does not define axis +/− directions.

1) First verify Robot No.1 and CONFIRMED = J/K 49138-20001. Check Jig A / Cavity 5 against the actual work. If these do not match, complete the approved Part-change procedure first; SELECTED alone is insufficient.

2) Open Main / Position Adjust, then use the image button. Locate number 33 in the top image. Cavity 5 has base 14800: 14800 + 33 = 14833, so use P14833. Do not enter 33 as the P number.

3) Close the image, enter 14833 in INPUT P NUMBER and tap search (⌕) or Enter. Verify P14833, Cavity 5 and JK_20001_L1 before touching an axis button. Record the current X/Y/Z values.

# 18 DENSO example: image point 33 → P14833 (continued)




J/K 49138-20001 reference: number 33 in the top image has a yellow line pointing to the left shoulder of the central hole. The right table gives Cavity 5 = 14800, hence P14833 for Cavity 5 / Jig A.

4) Press + or − for the intended axis on P14833. Each press changes the position value by 0.1 mm and submits an adjustment with no extra OK. Consecutive presses or multiple-axis changes are supported through the request queue, not simultaneous Controller execution of all axes.

5) After the intended presses, wait for the queue to finish and for CONFIRMED, which identifies the final request's P/axis/sign. Match returned X/Y/Z to Robot, Part, P and changed axes. Card values may update before completion, so an immediate visual change is not confirmation.

6) Inspect the workpiece under the WI and quality criteria. On Error, Timeout or mismatch, check status and readback before a new request; do not resend while the previous outcome is unknown. UNDO writes values back and also requires point/readback checks.

# 19 DENSO: Jig A and B offsets




Adjust the program offsets by Jig, Line and axis. This differs from editing an individual P in Main or shifting a range in Batch.

## 1. Match the Jig and Line to the job

Offset separates Jig A on the left and Jig B on the right, with Line and X/Y/Z rows in each column. These are offsets used by the program, not individual P coordinates from Main.

1) Check the Robot, Part, Jig, Line and axis against the job and the axis/Tool Line references before selecting a row.

2) Do not assume A and B must match or that an offset always affects every P. Follow how the job's program uses it.

Verify: The image's Line1 X values, Jig A=1.2 and Jig B=1.4, are separate fields-not recommended settings.

## 2. Use −/+ to change the current offset

− decreases and + increases the Offset of that Line for the selected axis and Jig by 0.1 mm per press. A write is requested without APPLY, unlike preparing Batch Delta.

1) Record the original value, check Jig, Line, axis and sign, then tap the intended field's −/+.

2) Offset does not use Main's repeated-press queue. While a request is busy, additional commands are not accepted; wait for confirmation and the returned value before the next press.

# 19 DENSO: Jig A and B offsets (continued)

Verify: The selected Line/axis changes by +0.1 mm or −0.1 mm per accepted press. Check the returned value after each request.

### 3. Type a new offset value

Editing the numeric field sets the new offset directly; it does not add a Delta. Committing the change or leaving the field sends set. Enter leaves the field, so there is no additional APPLY step.

1) Type only an approved final value. For example, if the intended new value is1.5, enter1.5-not merely the difference.

2) Check the sign, decimal and final value before leaving the field, then wait for the response.

Verify: The returned field value matches the approved final value. The1.5 example explains the command, not a recommended correction.

### 4. Verify the result and handle uncertainty

Match the Jig/Line/axis and before-and-after values, then follow the quality WI. A successful write does not prove the workpiece meets its quality criteria.

1) This page has no separate UNDO. Do not assume Main or Batch UNDO restores this offset field.

2) On Error, Timeout or a mismatch, stop, read the actual value and contact the responsible person. Restoring an old value is a new write requiring authorization.

Verify: Do not retry or enter0 to guess that the original value was restored. Confirm it from readback.

# 20 DENSO: Batch Shift




Shift the selected range of P positions by the same amount on one axis. Use it when a whole-range adjustment is approved, rather than changing points individually.

### 1. Check the job and unlock

Check the Robot, Part and work range first. LOCKED/UNLOCKED controls Batch editing, not machine readiness.

1) If LOCKED, tap the lock control four times in succession until UNLOCKED appears. If it fails, read the message rather than bypassing the condition.

Verify: UNLOCKED is shown before preparing the adjustment.

### 2. Select the Start / To range

The left field is the first P and the right field the last P. Position counts the selected points in the Part mapping. The image button opens the reference image.

1) Select both ends, then check the program labels and point count before changing a value.

Verify: The image example is 20101-20111, 11 points in CT_304_L2. Use the actual job range, not the example numbers.

### 3. Choose the axis and prepare Delta

X / Y / Z selects the axis. Delta is the signed change in mm. −/+ changes the prepared value by 0.1; it does not yet write the positions.

1) Use the axis and Tool Line Reference diagrams to select the approved axis and sign.

# 20 DENSO: Batch Shift (continued)

Verify: The selected axis and Delta match the job plan. The image value 0.1 is an example, not a recommendation.

### 4. APPLY and verify

APPLY starts writing the range adjustment; % shows progress, not distance.

1) Recheck the range, axis and Delta. Press APPLY once and wait for completion confirmation.

2) Check the returned position values against the intended range/axis, then follow the workplace quality-check procedure.

Verify: After completion, Delta and progress may return to 0. Judge the result from confirmation and read-back values, not 0% alone.

### 5. LAST / UNDO / lock again

LAST identifies the latest Batch. UNDO reverses a Handheld Batch for the same Part when its context is confirmed.

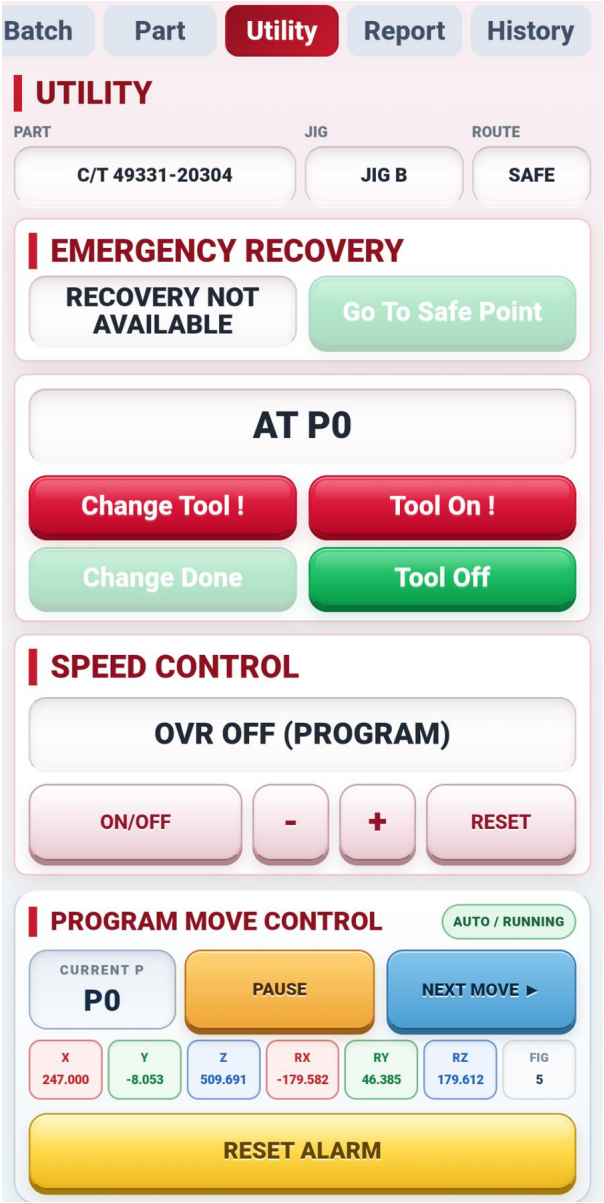
1) Read LAST before UNDO. If DIFFERENT/UNKNOWN SOURCE OR PART appears or the button is disabled, check the context rather than forcing an undo.

2) When finished and no operation is running, tap UNLOCKED to lock again.

Verify: After UNDO, verify the returned values again. Pressing the lock control during a job requests cancellation; it does not roll back values already written.

Caution: On Error, Timeout or RECOVERY REQUIRED, do not repeat APPLY. Have the responsible person check the actual state and values first.

# 21 DENSO Utility




Five utility groups cover Recovery, tool change/on-off, Speed, program control and Reset Alarm. Use only the group needed for an authorized task; they are not a sequence to run in order.

### 1. PART / JIG / ROUTE - identify the active job

PART is the selected part, JIG the work side, and ROUTE the route information. SAFE here is not permission to enter the machine area.

1) Check the Robot and these three fields against the job before using the controls below.

Verify: The displayed context matches the actual job. Otherwise, return to the Part/Jig selection first.

### 2. EMERGENCY RECOVERY - supported safe-point return

Go To Safe Point requests recovery for a supported Part. It is not a menu-navigation button.

1) Read the status first: RECOVERY NOT AVAILABLE or a disabled button means it is unavailable, even if ROUTE is SAFE.

2) If enabled, the authorized person follows the workplace recovery SOP and waits for the system result.

Verify: Verify the returned state/position under the SOP. A button press alone does not confirm recovery completion.

### 3. TOOL - tool positioning and on/off

Change Tool ! requests P191 from P0; Change Done requests P0 from P191. Tool On ! / Tool Off controls the tool through IO30; it does not isolate energy.

# 21 DENSO Utility (continued)

1) Before Change Tool / Change Done, confirm Tool Off and the correct starting position. If disabled or MOVING, wait and read the reason.
2) Under the approved tool-change procedure, press the required command once and wait for motion to finish; do not stack commands.
Verify: Verify the returned AT P191 or AT P0 and tool state against the request before continuing.

### 4. SPEED CONTROL - program speed or override

ON/OFF toggles override; −/+ changes its value by 5 percentage points. RESET disables override and sets its value to 30%; it is not RESET ALARM.
1) Read OVR OFF (PROGRAM) or the current override value first, then make only the approved change.
Verify: The readout shows the returned mode/value. OVR OFF uses program speed; it does not force running at 30%.

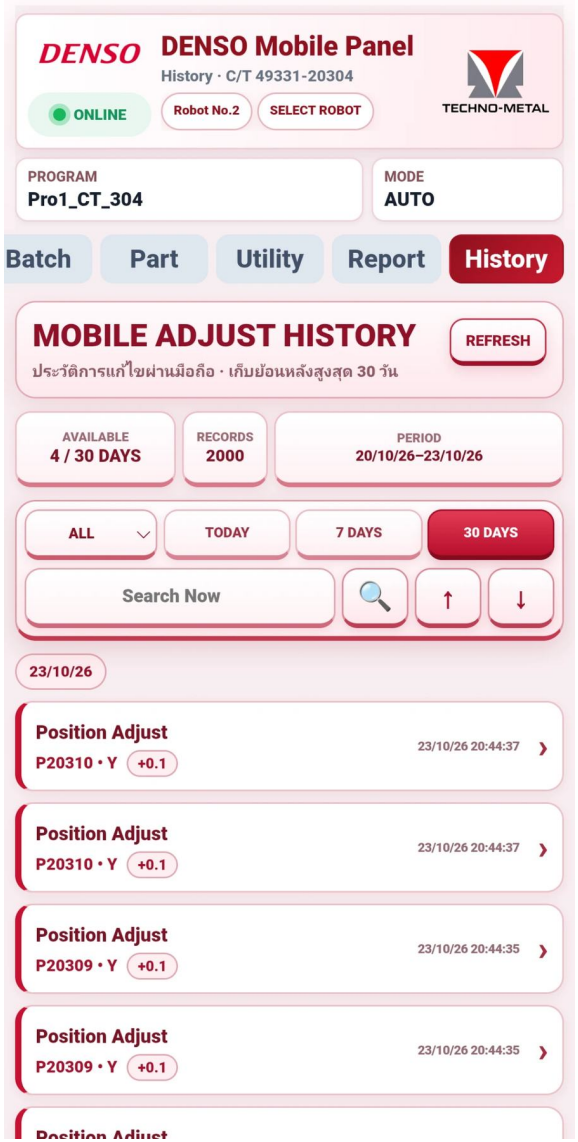
### 5. PROGRAM MOVE CONTROL - pause, continue or next movement

PAUSE requests a STEP STOP. CONTINUE is available from STEP STOP; HOLD is a different state and cannot use this CONTINUE control.
1) Read AUTO / RUNNING or the displayed state. Use only controls enabled by the system and authorized for the task.
2) NEXT MOVE requests the next movement. It may pass several non-motion instructions until actual position change is detected. Press once and wait for the result.
Verify: Check the returned program state and position; do not expect P to increase by one. STEP STOP=4 and HOLD=1 are different task-status codes.

### 6. CURRENT P / coordinates / RESET ALARM - read results and clear errors

CURRENT P identifies the current point. X/Y/Z show position, RX/RY/RZ orientation, and FIG the Robot configuration. These are readouts, not adjustment-entry fields.
1) Read and address the Alarm cause under the workplace procedure first. If authorized, press RESET ALARM once.
Verify: Wait for Controller confirmation and check the state again. If the fault remains, contact the responsible person instead of repeatedly resetting to continue.
Caution: Go To Safe Point, Change Tool, Change Done, CONTINUE and NEXT MOVE may move the machine. Follow the workplace SOP and authorization. PAUSE, Tool Off and RESET ALARM are not emergency-stop or energy-isolation devices.

# 22 DENSO adjustment history




DENSO adjustment history for Robot No.2 / C/T 49331-20304, with date filters and search.

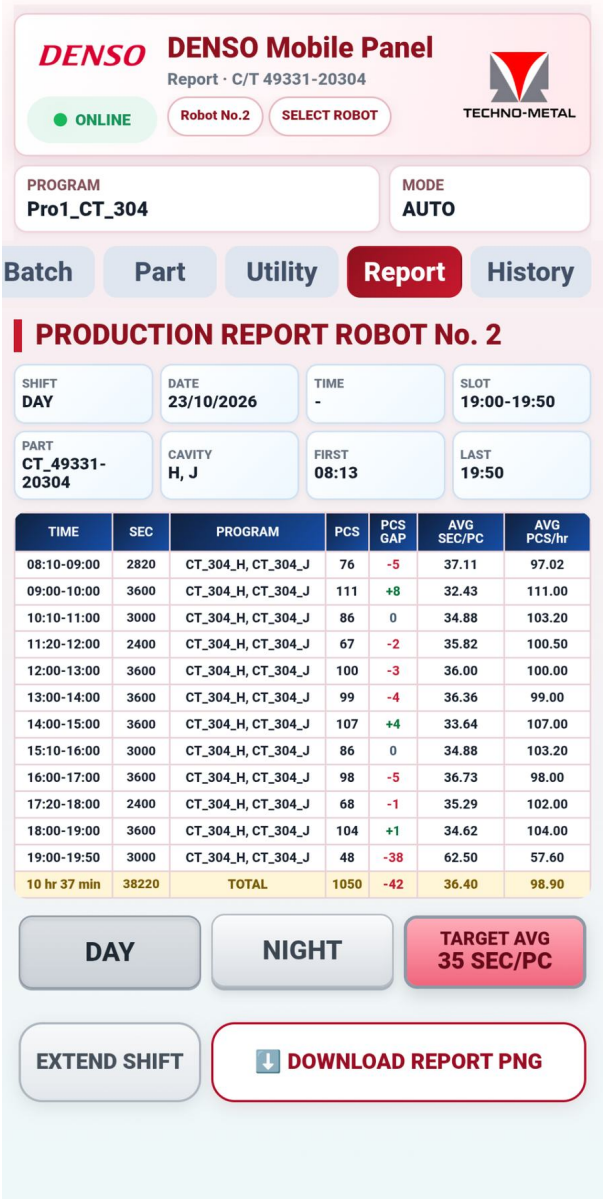
Find DENSO P-position history using date filters and search.

## Using and reading this screen

1) Check Robot No.2, C/T 49331-20304, Pro1_CT_304 and AUTO before interpreting the history.

2) AVAILABLE 4/30 DAYS, RECORDS 2000 and the period 20/10/26-23/10/26 describe the available scope. Use filters and Search Now.

# 23 DENSO: production report




Reports are scoped by shift and date. Read the part inside the report, not just the current app-header part.

## Using and reading this screen

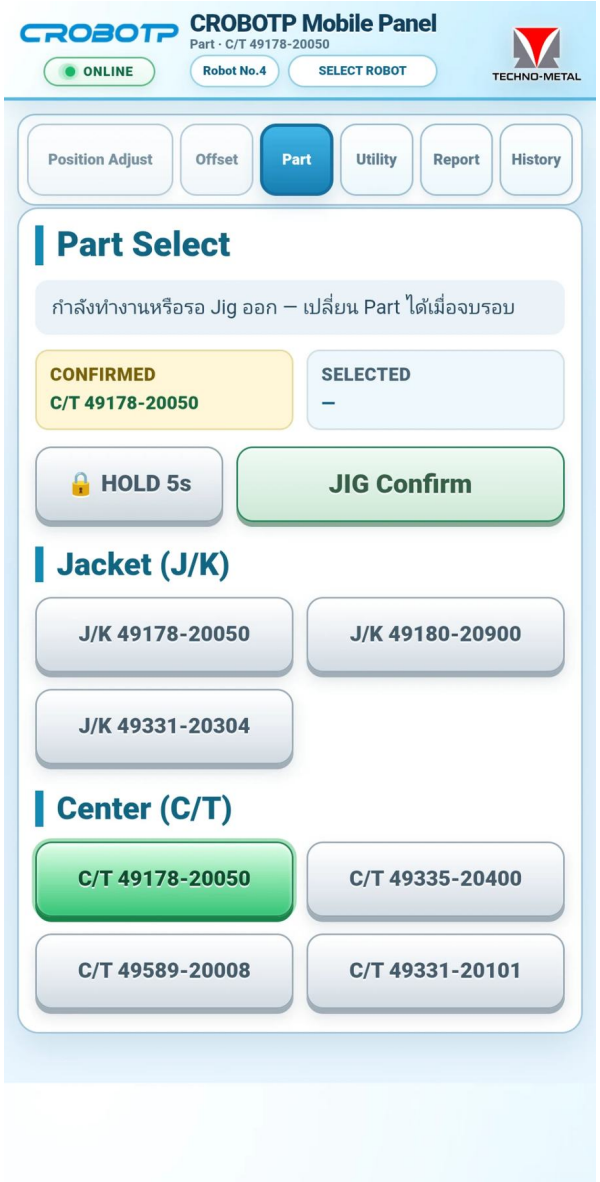
1) Check report Robot No.2, DAY and 23/10/2026, then PART, CAVITY, FIRST and LAST.

2) The table separates TIME, SEC, PROGRAM, PCS, PCS GAP and AVG. The shown total is 1,050 PCS, 36.40 SEC/PC, with target 35 SEC/PC.

3) DAY/NIGHT changes shift view; DOWNLOAD REPORT PNG exports a report image. This manual's PDF button is a different download.

Caution: Rows include H/J together; do not halve PCS between programs.

# 24 CROBOTP part selection




CROBOTP uses GR values as offsets for LP points, rather than editing the original point coordinates as DENSO Main does for P points. Because GR storage is limited, different parts reuse register fields. When changing over to a new part, its position-offset GR values start from zero so the previous job's offsets do not carry into the new job.

## Using and reading this screen

1) CONFIRMED is accepted; SELECTED is pending. Choose the full code within Jacket or Center.

2) HOLD5s requests server-authorized unlock; early release cancels. The UI selection window is55seconds, not a five-second robot stop.

3) Select the part for the job, check that the Part and Jig match the work, then press JIG Confirm. Read Part and Robot in the dialog, check that the jig matches the work, then confirm. Wait for the system to accept it and for CONFIRMED to show the correct part. Selecting a part name alone does not start the program.

4) After changeover and preparation of the new part are complete, check its offsets using fresh status. To reuse saved adjustments, open Offset Preset for that Robot/new Part: use ON for the latest set or Recall for a selected historical set, then wait and verify the returned values. Selecting a part name does not mean its preset has been recalled.

Caution: Cycle completion and readiness are required. On timeout or identity mismatch, recheck status; do not bypass the lock.

# 25 CROBOTP: Position Adjust




Robot No.4 Position Adjust, C/T49178-20050, Jig A: LP1 and its X/Y/Z mapping to GR101/102/103.

LP identifies a robot work point: LP33 is point 33 in the reference image. The X/Y/Z numbers on its card are offsets applied to that point, not its original coordinates.

## Using and reading this screen

1) Check the confirmed Robot and Part, choose the correct jig and enter the LP number in INPUT LP NUMBER. Searching displays a point card; it does not move the robot.

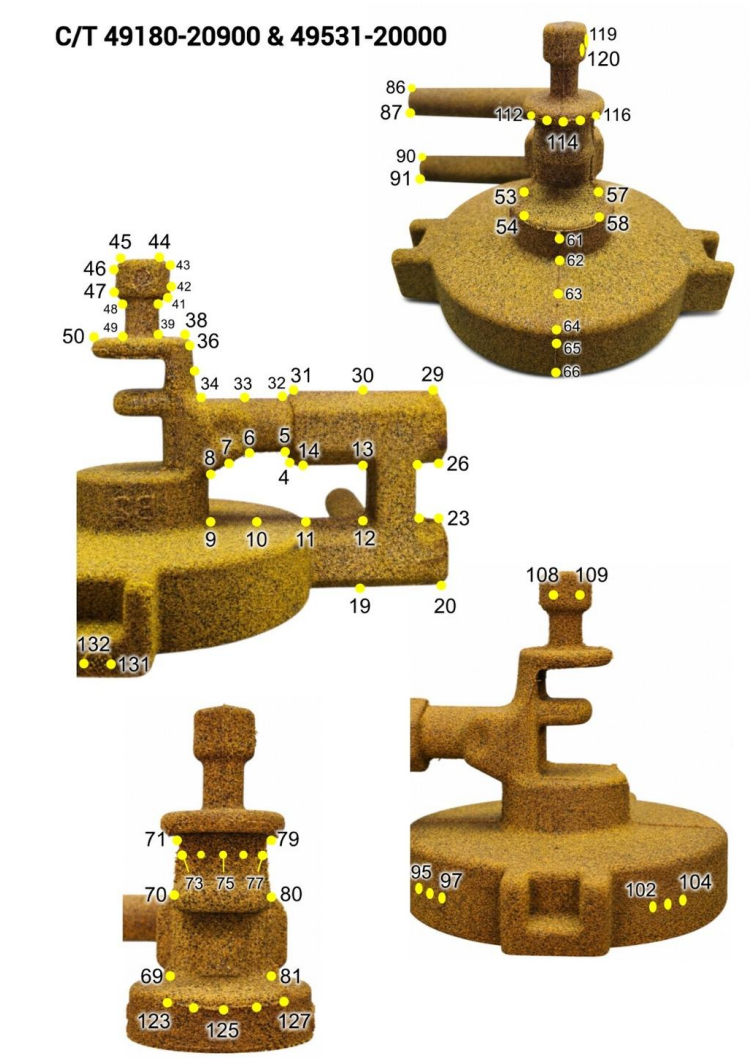
2) In this example, LP1 on Jig A belongs to L2: X uses GR101, Y uses GR102 and Z uses GR103. Each GR is the storage field for that axis offset. Read and use −/+ in the intended axis field.

3) Each + press adds 0.1 mm to the offset; each − press subtracts 0.1 mm. The increment is fixed at 0.1 mm, not selectable. Individual point buttons require no additional APPLY.

4) Wait for the response, then check the LP, jig and displayed offsets against the intended adjustment. If the outcome is unclear, check current values before repeating a press.

Caution: GR numbers and LP ranges depend on the part and jig. Do not reuse the example GR101/102/103 for every point. Adjust only approved points and axes, and inspect the workpiece under the WI before continuing production.

# 26 CROBOTP example: image point 33 → LP33 / Jig A




C/T 49180-20900 / 49531-20000: point 33 is on the upper edge of the horizontal arm between 34 and 32. Search LP33 directly, without adding a DENSO-style base.

Read the LP number and verify Jig, Line and GR before checking the offset result.

Caution: Obtain approval for the Robot, Part, jig/cavity, point and axis, and check machine condition under the WI. Stop and check stale or mismatched data. The part image does not establish the axis +/− direction.

1) Check Robot No.3 and CONFIRMED Part C/T 49180-20900 / 49531-20000 against the workpiece. This example uses Jig A / Cavity 1. If it is only SELECTED, complete part/jig confirmation first.

2) Open Position Adjust, select JIG A and find point 33 on the horizontal arm in the image. This is LP33 in L2 (LP1-67); do not add a DENSO-style base.

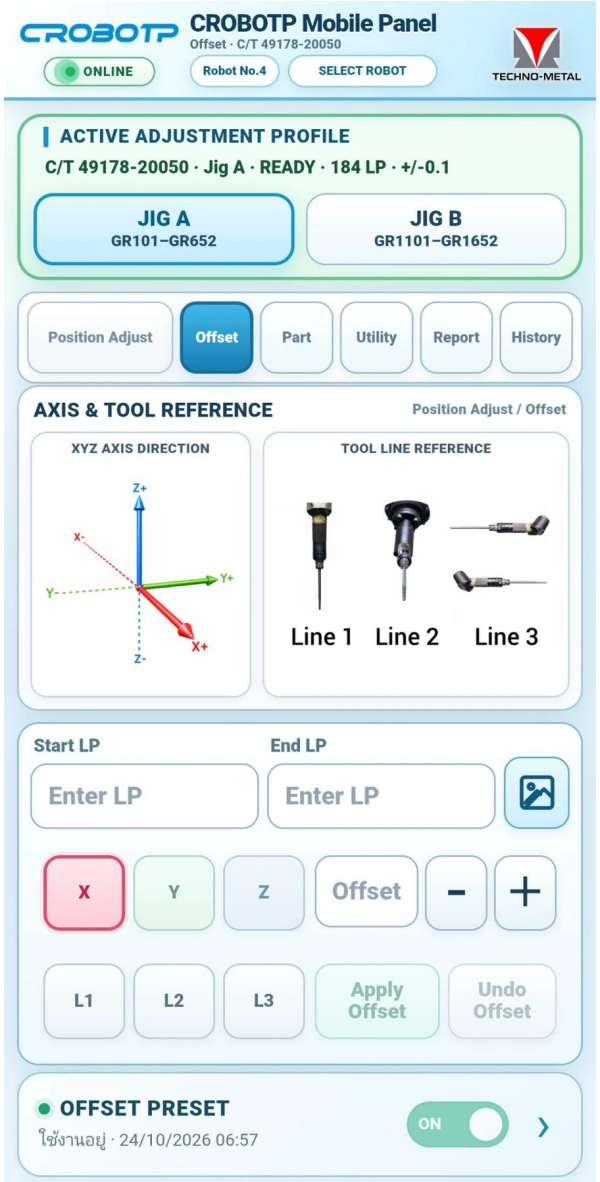
3) Close the image, enter 33 in INPUT LP NUMBER and search. Check LP33 · Jig A · L2: X uses GR197, Y GR198 and Z GR199. Read the offsets before adjusting.

4) Press the intended axis's + to add 0.1 mm or − to subtract 0.1 mm. The increment is fixed; no other Step is selectable. Individual point buttons write immediately, with no additional APPLY.

5) Wait for the response and read LP33 / Jig A again. For example, if X starts at 0.0, one successful + press should result in X = 0.1 mm. If the value differs or the operation is still pending, do not press again to guess the outcome.

6) Inspect the workpiece under the WI before continuing production. For Undo, verify the same LP33 / Jig A and check X/Y/Z after restoration. On timeout or an unknown result, check current values and contact the responsible person before continuing.

# 27 CROBOTP: Line and LP-range Offset




Adjust offsets as a group: either the selected line offset or several points in an LP range. Check which mode is active before pressing Apply.

## Using and reading this screen

1) Match the Robot/Part to the job, then choose Jig A or B and the X, Y or Z axis.

2) Line mode: select L1, L2 or L3 to change that line's offset. Start/End LP does not limit the adjustment in this mode.

3) LP-range mode: deselect the line, set Start LP and End LP, and check that all points in the range have loaded.

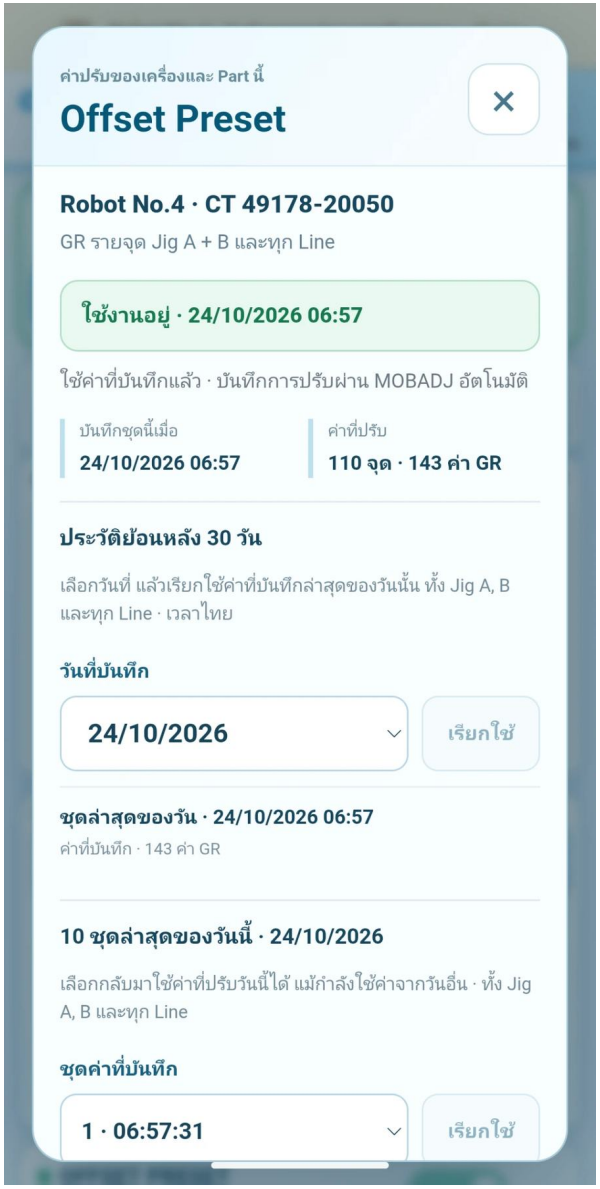
4) Set the signed adjustment and press Apply. Check the confirmation for the intended jig/line or LP range before confirming. Wait for the result and inspect returned values.

5) Undo restores the X/Y/Z values saved before Apply for the same line or LP range, not just the currently selected axis. Check the scope in the confirmation before using it.

6) OFFSET PRESET stores a part's saved values: ON uses the latest set; OFF zeros the part's GR while keeping saved sets; Recall applies the historical set you selected. See Offset Preset for details.

Caution: Choose the mode deliberately: line and LP-range adjustments affect different scopes. Narrowing Start/End LP while a line remains selected does not restrict the line adjustment to those points. Check current values before repeating an unclear operation.

# 28 CROBOTP: Offset Preset




Offset Preset saves adjustments by Robot and Part, so a job's values can be restored after a part changeover that starts position-offset GR at zero. This avoids remembering and re-entering every offset, but the selected set must match the part actually in use.

## Using and reading this screen

1) Check the Robot/Part and timestamp. A set covers both jigs and all lines of that part, not only the jig/line currently displayed.

2) ON applies the latest saved set; it need not be the historical set currently selected in the dialog.

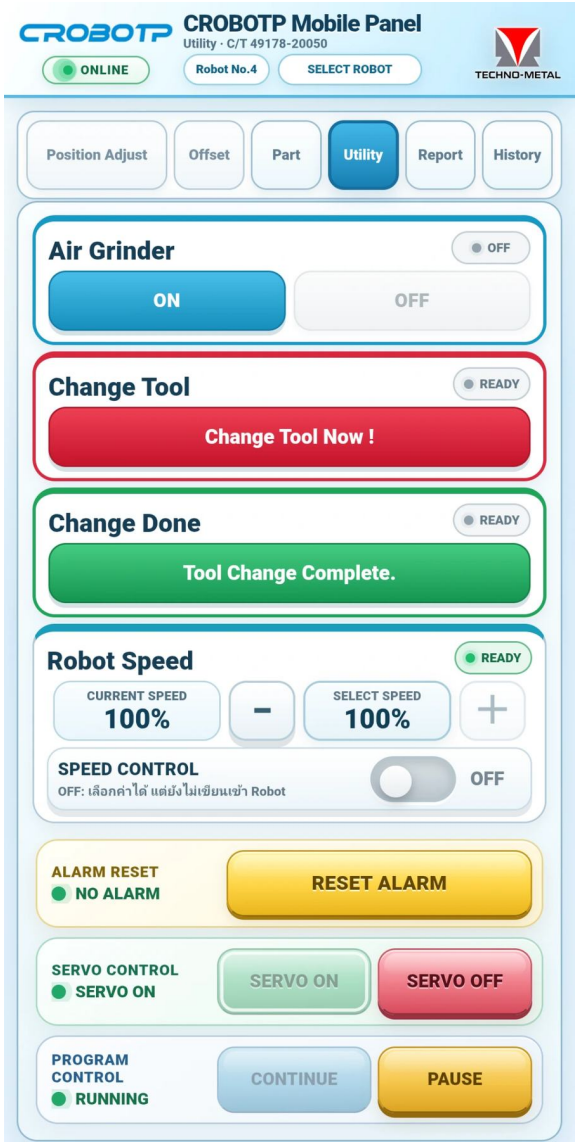
3) OFF zeros the part's GR but keeps saved sets. It does not simply hide a label or disable saving.

4) Recall applies your selected historical set. Choose a date from the 30-day history for that day's latest set, or a time from today's 10 sets, then use the Recall button beside that selection.

5) Read the confirmation before ON/OFF/Recall. Wait for success and verify the returned values match the intended set.

Caution: A preset changes values across the part. Verify the job and set before confirming. If the result is unclear, check status and current values before repeating.

# 29 CROBOTP Utility




Grinder OFF: the tool-change fields show READY in this image.

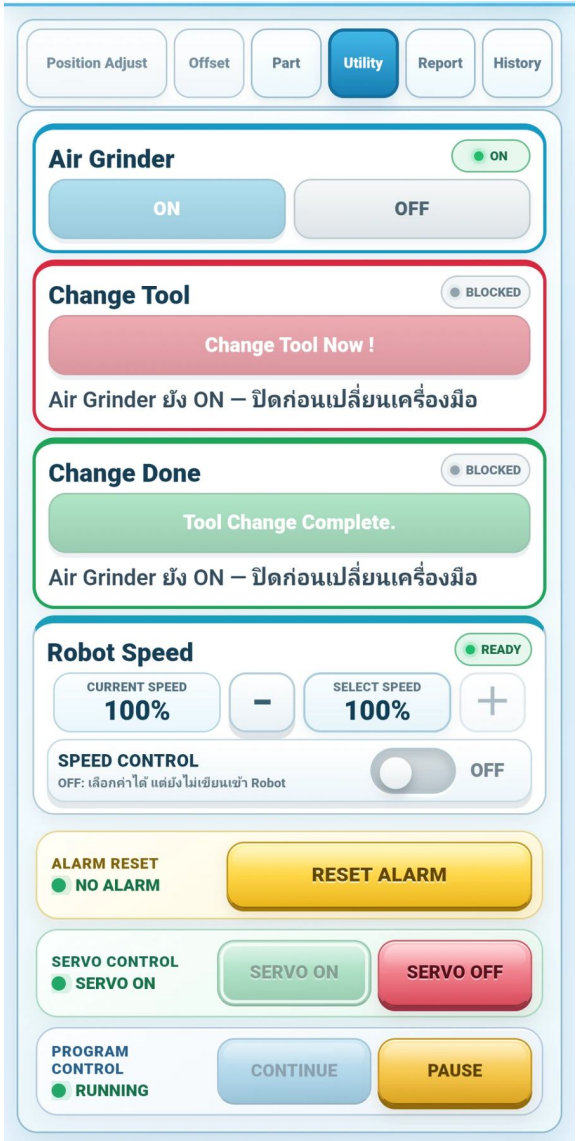
This page groups grinder, tool-change, speed, alarm, servo and program controls. The two images show the same page with the grinder OFF and ON.

## Using and reading this screen

1) Check the Robot/Part and read the Air Grinder ON/OFF badge. ON/OFF commands the grinder; it is not merely a display switch.

2) With Grinder ON, Change Tool and Change Done are BLOCKED. Switch the grinder off under the site procedure, then check the conditions and READY state again.

# 29 CROBOTP Utility (continued)




Grinder ON: the tool-change fields show BLOCKED.

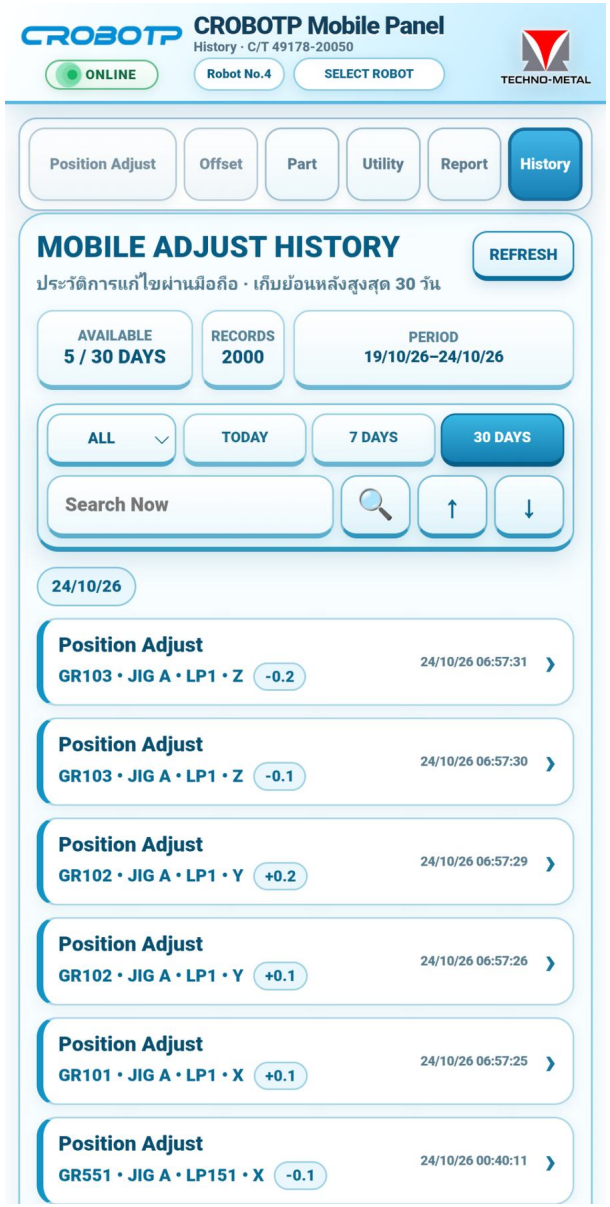
3) Change Tool Now ! and Tool Change Complete. are different commands. Follow the approved tool-change sequence; READY is not permission to touch the machine.

4) CURRENT SPEED is the current value; SELECT SPEED is the chosen value. With SPEED CONTROL OFF, −/+ changes only the selection. ON applies it and subsequent −/+ changes immediately; switching OFF restores the pre-enable speed.

5) The lower controls separate RESET ALARM, SERVO ON/OFF and CONTINUE/PAUSE. Read each state before use under the WI: grinder state does not establish that the servo or program has stopped.

Caution: Grinder OFF/READY is not immediate permission for tool handling; follow site safety procedures.

# 30 CROBOTP adjustment history




Read entries by timestamp, point, jig, axis and GR; do not assume all historical entries remain active.

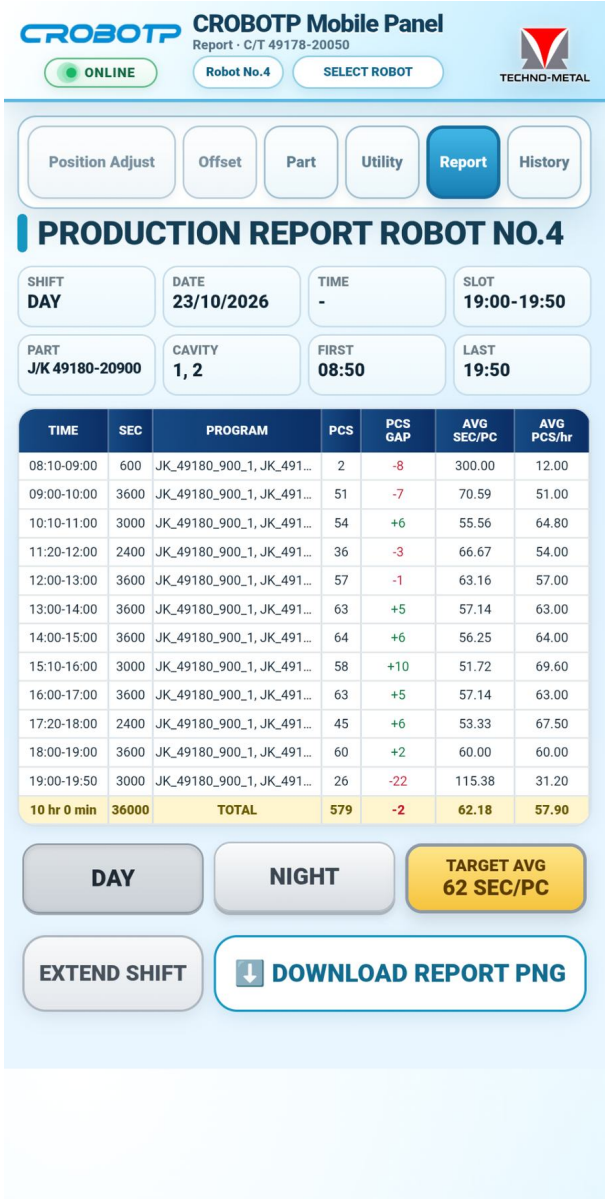
## Using and reading this screen

1) Read AVAILABLE5/30 DAYS together with PERIOD and active filters to understand the displayed history scope.

2) ALL filters category; TODAY/7DAYS/30DAYS filter dates. Search Now searches; REFRESH updates entries.

3) The GR103 · JIG A · LP1 · Z entry at 06:57:31 shows -0.2. Read it with preceding entries; do not sum values without knowing the record semantics.

# 31 CROBOTP: production report




Reports are scoped by shift and date. Read the part inside the report, not just the current app-header part.

## Using and reading this screen

1) Check report Robot No.4, DAY and 23/10/2026, then PART, CAVITY, FIRST and LAST.

2) The table separates TIME, SEC, PROGRAM, PCS, PCS GAP and AVG. The shown total is 579 PCS, 62.18 SEC/PC, with target 62 SEC/PC.

3) DAY/NIGHT changes shift view; DOWNLOAD REPORT PNG exports a report image. This manual's PDF button is a different download.

Caution: Historical report part J/K49180-20900 may differ from current header C/T; do not force them to match.

# 32 Terms and metrics

### P / LP / GR

P identifies a DENSO position, such as P20101. LP identifies a CROBOTP position, such as LP1. GR identifies the register shown against an axis; LP1 Jig A shows X=GR101, Y=GR102, Z=GR103. These identifiers are not interchangeable.

### Part / Program / Jig / Cavity / Line

Part is the part identifier; Program is the displayed program name; Jig is the A/B fixture. Cavity is the number or letter marked on the workpiece, such as H, J or 1, 2. Match the workpiece marking to the displayed Cavity. Line is the L1/L2/L3 grouping shown with tool references. Cavity is not a slot or position number on the Jig.

### PCS / SEC / SEC/PC / PCS/hr

PCS is quantity, SEC is seconds, SEC/PC is time per piece, and PCS/hr is pieces per hour. The CROBOTP report example gives 36,000÷579≈62.18 SEC/PC and 579×3,600÷36,000=57.90 PCS/hr. These explain the screenshot, not recommended settings.

### 0 / -- / No data / No target

0 PCS is a displayed zero. -- means no value is shown in that field. Missing data and missing target are different conditions. Do not replace all of them with zero or diagnose a machine failure from one chart cell.

### PCS GAP

Actual pieces − (recorded production seconds ÷ Target AVG): + is above target, − below. Calculate per Part, add, then round to whole pieces; a small difference may display0. - means insufficient data, not zero or scrap. This duration is not the scheduled work-time denominator of SHIFT SCORE. See Graphs for examples.

### Target colors

Server-reported status is preferred. Numeric fallback: FAST more than1second faster than target; ON TARGET actual≤target; SLIGHTLY SLOW less than1second slower; SLOW at least1second slower. Missing usable target/average becomes NO TARGET/NO DATA.

### Ranking

Check data status, then SHIFT SCORE. Within a group, higher displayed one-decimal score comes first; ties use higher whole-shift PCS, then lower Robot number. Positive output always precedes zero/unknown output. Output with score - follows scored Robots in its group. ONLINE/STALE with usable Reports receive rank numbers; OFFLINE and WAITING/CHECK REPORT are separate unranked groups. See Ranking for grouping details.

## 33 Simulator

Open Simulator in the HTML guide to select a Robot and practise its adjustment screen. This PDF provides reading guidance.